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DOI: 10.1002/jpen.2540

ORIGINAL RESEARCH

Assessment of body composition in pediatric intestinal failure: A comparison study

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Funding information
Rare Disease Foundation

Abstract

Background: The objective of the study was to compare bioelectrical impedance analysis (BIA) and skinfolds with dual energy x-ray absorptiometry (DXA) in the assessment of body composition of children with intestinal failure. DXA is the reference method for body composition assessment in clinical settings.

Methods: Children aged 1–18 years with intestinal failure whohave DXA as part of routine clinical monitoring were eligible. BIA measured total body water on the same day as DXA. Skinfold measurements were taken at four sites: triceps, biceps, subscapular, and suprailiac. Percentage of fat mass (%FM) and fat-free mass (%FFM) were derived from resistance and reactance measured by BIA by using age-specific equations. Percentage of FM was calculated from skinfold measures by using age-specific equations. Data on patient characteristics, intestinal failure–related factors, and feeding method were collected. Paired *t* test examined differences in %FM and %FFM and Bland-Altman analysis determined the agreement between BIA, skinfolds, and DXA. Marginal linear model assessed the effect of age, sex, and feeding method on the difference in body composition obtained between DXA and BIA and between DXA and skinfolds.

Results: Sixty-eight children with intestinal failure, mean age 8.9 ± 4.2 years, were studied. There was no difference between %FFM and %FM obtained by DXA and BIA ($P = 0.26$), with a mean bias (95% CI) of −0.69 (−1.9 to 0.5) for %FFM. Sex and age were individually and jointly associated with the bias observed between DXA and BIA ($P < 0.05$). Skinfold and DXA measurements were significantly different ($P < 0.05$).

This study was registered at ClinicalTrials.gov (registration number: 1000069821).

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Conclusions: BIA is an acceptable clinical tool for assessing body composition in pediatric intestinal failure.

KEYWORDS
bioelectrical impedance analysis, body composition, intestinal failure, nutrition assessment, pediatrics

INTRODUCTION

Over the last few decades, improvement in clinical care has resulted in longer life expectancy of children with intestinal failure (IF).[1,2] With improved survival, additional complications have emerged.[3–6] Recent data indicate that the body composition of children with IF is abnormal with a relatively high fat mass (FM) and low fat-free mass (FFM) when compared with healthy children.[7–11] Most concerning is that abnormal body composition is present in both normal and abnormally growing children with IF based on weight-for-age (WFA) and height-for-age (HFA) *z* scores.[10] This emphasizes that growth charts have poor sensitivity for monitoring response to treatment.[12] Abnormal body composition has been linked to poor prognosis, increased length of hospital stay, and decreased strength, as well as an increased risk of other noncommunicable diseases, such as diabetes and cardiovascular disease, in adulthood.[13–15] Therefore, in addition to height and weight, FM and FFM should also be measured as part of routine nutrition assessment in the pediatric population. This is relevant because childhood is considered the incubation period for chronic diseases later in life.[16,17]

The gold standard for measuring body composition is the four-compartment model, which is not available clinically because of its need for specialized equipment and invasive procedures.[18] Dual energy x-ray absorptiometry (DXA) provides a rapid, precise, and noninvasive measure of FM, FFM, and bone mineral density and is regarded as the reference method to measure body composition in the clinical setting.[19] However, its application for routine clinical use is precluded by its relatively high cost and restricted health insurance coverage. Another option is four-point skinfold thickness measurements, which provide an estimate of total body fat.[20] However, despite being inexpensive and noninvasive, there are several possible sources of error in this method, and thus it may not be reliable for routine use, particularly in younger children.[21]

Bioelectrical impedance analysis (BIA) is a simple to use, noninvasive, and inexpensive method, which makes it a useful technique for measuring body composition in the clinical setting.[22] Importantly, BIA can be used at the bedside, making it a convenient tool to facilitate the tailoring of nutrition plans to the specific needs of each patient.[23] BIA measures total body water through the resistance of the body to a small alternating current, therefore controlling for fluid intake is of utmost importance.[24] A previous study assessing body composition in pediatric IF using BIA did not control for fluid intake.[25] In this study, BIA was compared with deuterium dilution; however, in children with IF, the absorption of enterally administered deuterium isotope could be variable. There is a need to evaluate the appropriateness of using BIA to assess body composition in pediatric IF while controlling for fluid intake (both enteral and intravenous). The ability to assess body composition in children with IF on a routine basis as part of standard care would allow for a more tailored approach that could help optimize not just height and weight but overall body composition and health.

Therefore, the primary objective of this study is to compare results of body composition obtained by BIA against DXA to determine if BIA can be used as a tool for clinical assessment/monitoring of body composition in children with IF. We hypothesized that BIA is equivalent to DXA for assessing body composition in pediatric patients with IF. As a secondary objective, the four-point skinfold method was also compared with DXA. We hypothesized that the four-point skinfold method is inferior to DXA for assessing body composition in pediatric patients with IF.

METHODS

Study population

This was a comparison study of body composition assessment methods in children aged 1–18 years with IF managed by the intestinal rehabilitation program at SickKids Hospital in Toronto, Canada. This included patients who were receiving parenteral nutrition (PN) (exclusively and/or partially) as well as those who had weaned from PN and were fully enterally fed. Children who were having their annual DXA scan as part of routine clinical monitoring were eligible. Exclusion criteria were set to patients who used a wheelchair or who were nonweight bearing, those with clinically evident shifts in fluid compartments (eg, edema or ascites), and those with severe developmental delay or unwilling to participate. All participants underwent assessments of body composition on the same day as their clinic appointment. Body composition assessment was completed using BIA, four-point skinfold thickness measures, and DXA. A gift certificate was provided to each child who participated.

Ethical statement

The Research Ethics Board of the Hospital for Sick Children approved the study. Consent was obtained from the parents of

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each child and assent was obtained from individual patients. This study was registered at ClinicalTrials.gov (registration number: 1000069821).

Demographics and nutrition data

The following data were collected from the medical charts: FM, lean body mass, and bone mineral content as measured by DXA (on the same day the BIA and skinfolds measures were obtained), IF diagnosis, gestational age, current age, age at onset of PN, duration of PN, and remnant intestinal anatomy. Remnant intestinal anatomy was determined based on surgical measurements at the time of the last surgical intervention.[26] A breakdown of nutrition intake (percentage of enteral nutrition [EN] and percentage of PN energy) was recorded at the time of the body composition assessment. This was collected as a percentage of the overall energy intake (PN energy/[total daily energy intake EN + PN] × 100%), as well as a PN dependency index (PNDI). The PNDI was calculated as the ratio of the nonprotein energy provided by PN over the resting energy expenditure calculated by using the Schofield equation.[27] Children with a PNDI value of ≥120% were considered highly dependent on PN, whereas those with a value of <120% were moderately dependent.[27] All patients in this study received some amount of EN. The degree of tolerance/absorption of EN by fecal fat test was not assessed in this study. Patients who had a PNDI of >80% demonstrated greater dependence on PN. As part of clinical monitoring, serum creatinine and urea were collected and reported as markers of hydration status on the day of BIA measurement.

Anthropometry

Weight and height were collected at the time of the body composition measurements and converted to sex-specific WFA and HFA *z* scores using the World Health Organization data set.[28] Heights and weights were measured using standard equipment, including a clinical calibrated electronic scale and a wall-mounted stadiometer.

DXA

All cases underwent assessment using DXA (GE-Lunar DPS Pro [GE Healthcare]) in the radiology department of SickKids as part of routine annual clinical monitoring. DXA was performed with the patient wearing light, metal-free clothing. Radiation exposure was <2 microSv/scan.[29] The scan measured total FM, lean body mass, and bone mineral content. Output data were reported in gram amounts as well as percentage. Percentage of FFM was determined using the sum of lean body mass and bone mineral content divided by 100%.

BIA

Whole-body BIA was taken on the right side of the body using a multifrequency analyzer BIA, model 101A (RJL Systems). BIA measurements were obtained with patients in the supine position. Current-injector electrodes were placed just below the phalangeal-metacarpal joint in the middle of the dorsal side of the right hand and below the metatarsal arch on the superior side of the right foot. Detector electrodes were placed on the posterior side of the right wrist, midline to the pisiform bone of the medial side. Three readings of resistance and reactance were taken from each patient, and the mean was reported.

To control for the effect of hydration on BIA, all patients receiving any amount of EN underwent a 10-h overnight fast. Patients receiving PN (exclusively and/or partially) received their prescribed PN fluid volume based on fluid requirements to ensure appropriate fluid balance, hydration status, and growth. To control for the effect of intravenous infusions on BIA, patients receiving PN were cycled off PN for a minimum of 4 h before measurement of BIA. All patients were asked to void the bladder before the test. Those who have a stoma were asked to empty their appliance before the test. Patients were instructed to avoid exercise for at least 2 h before the test.

The undisclosed proprietary equation developed by the manufacturer of the BIA model was not used in this study because the assumptions used in this equation are related to adult patients. Specific pediatric equations from Fjeld et al were used to determine total body water using the impedance (Z) at 50 kHz, body weight and height.[30] The equation is shown below.[30] Impedance is calculated as the sum of resistance squared and reactance squared. FFM was then determined under the assumption that the hydration of FFM is 77%.[31]

$$\text{Total body water} = 0.18\ \text{height}^2/Z + 0.39\ \text{weight} + 0.76$$

Skinfolds

Four-point skinfold thicknesses (triceps, biceps, subscapular, and suprailiac) were measured in triplicate to the nearest 0.1 mm with a skinfold caliper (Holtain LTD) to obtain estimates of FM. The mean values were recorded. Nearly all measures were completed by one specified clinician to maintain accuracy and consistency among assessment. There were three measures completed by another trained clinician; however, a prior interrater variability study demonstrated a difference of <5% between the two clinicians. These skinfold measures were then used in an equation from Durnin and Womersley to derive body density.[32] Percentage of FM was then calculating using the Siri (1956) equation[33] and %FFM was calculated by difference based on body weight.

Statistical analysis

Paired *t* test was used to compare BIA against DXA as well as skinfolds against DXA. Bland-Altman assessed agreement between

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BIA and DXA as well as skinfolds and DXA. A marginal linear model was used to assess the effect of age, sex, and feeding method (enteral/parenteral) on the difference between body composition assessment methods. A linear model was fit by treating the difference between BIA and DXA (%FFM/%FM and absolute values in kilograms FFM/FM) as the response variable and sex as a covariate. A joint linear regression model was then used to control for the combined effect of age and sex. Kolmogorov–Smirnov test was used to test for normality. Significance was set at a criterion of $P \leq 0.05$. All analysis was performed by R-4.2.2 for Windows.

RESULTS

The data were found to be normally distributed and are presented as means and SDs.

Patient characteristics

Sixty-eight children with IF, mean (SD) age 8.9 (4.2) years and 46 (67.6%) of whom were male, were enrolled over a 2-year period between November 2020 and December 2022. Patient characteristics are reported in Table 1. Twenty-five (36.8%) patients were dependent on PN. The etiology of IF included short bowel syndrom secondary to gastroschisis (13.2%), volvulus (14.7%), necrotizing enterocolitis (22.1%), intestinal atresia (22.1%), and Hirschprung disease (7.4%). Motility disorders and congenital enteropathies represented 13.2% and 2.9%, respectively. Serum urea and creatinine were within reference range for all patients, suggesting adequate hydration status. All 68 participants underwent body composition assessment.

Anthropometry

The cohort demonstrated normal anthropometric growth, with mean WFA and HFA *z* scores of −0.1. There were no significant differences in growth characteristics between male and female patients or those patients receiving PN vs patients who were fully enterally fed (Table 1).

Comparison of body composition assessment methods

The body composition measurements by the three different techniques are summarized in Table 2 (%FFM/%FM) and Table 3 (FFM/FM kg). Paired *t* test revealed that the results for FM and FFM obtained by BIA were not different from those obtained by DXA (percentage $P = 0.26$; kilograms $P = 0.59$). Bland-Altman analyses of body composition measurements (%) by BIA and DXA technique showed a mean bias (95% CI) of 0.69 (−0.5 to 1.9) and −0.69 (−1.9 to 0.5), for %FM (Figure 1A) and %FFM (Figure 1B), respectively. Bland-Altman analyses of body composition measurements (kilograms) by BIA and DXA technique showed a mean bias (95% CI) of 0.14 (−0.4 to 0.6) kg and −0.13 (−0.6 to 0.4) kg for FM (kilograms) (Figure 1C) and FFM (kilograms) (Figure 1D), respectively. Of the 68 participants, 67 (98.5%) fell within 10% of the mean bias. The %FFM and %FM obtained from skinfolds were significantly different from that obtained from DXA ($P = 0.001$) (Table 2).

Effect of age, sex, and method of feeding

Marginal linear mixed model revealed a significant but weak association (based on $r^2 = 0.06$, $P = 0.04$) between age and the difference between DXA and BIA for both %FFM and %FM. Regression equation for %FM ($y = 83.73 + -0.33$ [age]) is shown in Figure 2A and for %FFM ($y = -3.73 + 0.33$ [age]) in Figure 2B. Sex was also found to be significantly associated with the difference between DXA and BIA for both %FM and %FFM ($r^2 = 0.14$, $P = 0.002$). The regression equations for %FM and %FFM are $y = 7.27 + -3.94$ (sex) and $y = -7.17 + 3.94$ (sex), respectively. Joint linear regression model to test the combined effect of age and sex revealed both together had a significant effect on the difference between DXA and BIA for both %FM and %FFM ($r^2 = 0.32$, $P = 0.02$). The linear relationship for %FM and %FFM are $y = -0.05 + -0.26$ (sex × age) + −0.44 (sex) + −0.39 (age) and $y = 0.05 + 0.26$(sex × age) + 0.44 (sex) + 0.39 (age), respectively. Mode of feeding (parenteral [exclusive and/or partial] vs EN) did not have a significant effect on the results.

DISCUSSION

This is the largest study to our knowledge comparing body composition assessment between BIA and DXA in children with IF. The results demonstrate that FM and FFM measured by BIA in our cohort of patients with IF were not different from those measured by DXA. Indeed, the mean biases obtained by Bland-Altman analysis of agreement between these techniques were small with narrow 95% CI, suggesting that BIA may be suitable for assessment of body composition in pediatric IF. When controlling for the effect of confounders, we found that mode of feeding (parenteral [exclusive and/or partial] vs EN) did not influence the results. However, the difference between DXA and BIA was affected by age and sex, both individually and jointly. In older children, DXA gives a higher FFM and lower FM than BIA. This was demonstrated in the Bland-Altman plot in which the mean bias between DXA and BIA becomes less with increasing age and values of FFM and FM. In addition, the difference between DXA and BIA was greater in male patients than in female patients. When age and sex were combined in the model, the difference between DXA and BIA remained positive in favor of DXA for FFM and negative for FM. Although both sex and age contribute to the introduction of bias in the results, only one patient had a difference, which was >10% of the difference.

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TABLE 1 Characteristics of pediatric patients with intestinal failure.

	All patients (*n* = 68)	Receiving PN (*n* = 25) (36.8%)	Weaned off PN (*n* = 43) (63.2%)
Male sex, *n* (%)	46 (67.6)	19 (76.0)	27 (62.8)
Gestational age, mean weeks (SD)	34.2 (4.1)	35.6 (3.5)	33.4 (4.2)
Age at DXA, mean years (SD)	8.9 (4.2)	8.5 (4.0)	9.2 (4.2)
Diagnosis, *n* (%)			
Gastroschisis	9 (13.2)	6 (24.0)	3 (7.0)
Necrotizing enterocolitis	15 (22.1)	2 (8.0)	13 (30.2)
Hirschsprung disease	5 (7.4)	3 (12.0)	2 (4.7)
Volvulus	10 (14.7)	2 (8.0)	8 (18.6)
Intestinal atresia	15 (22.1)	6 (24.0)	9 (20.8)
Congenital enteropathy	2 (2.9)	2 (8.0)	0 (0.0)
Dysmotility	9 (13.2)	3 (12.0)	6 (14.0)
Other	3 (4.4)	1 (4.0)	2 (4.7)
Remaining small bowel length, mean % (SD)	49.2 (34.7)	33.0 (32.4)	58.8 (32.7)
Remaining large bowel length, mean % (SD)	74.1 (40.1)	56.0 (36.0)	84.6 (38.9)
Age at PN initiation, mean days (SD)	208.2 (602.1)	136.5 (395.2)	251.9 (700.2)
Duration of PN until achieving enteral autonomy, mean years (SD)	–	–	1.8 (2.5)
Duration of PN at the time of the study, mean years (SD)		7.4 (3.6)	–
PNDI	–		–
PNDI >120%, *n* (%)		7 (28.0)	
PNDI 120%–80%, *n* (%)		9 (36.0)	
PNDI <80%, *n* (%)		9 (36.0)	
PNDI, mean % (SD)		86.3 (45.4)	
PN energy based on total daily energy intake (EN + PN), mean % (SD)		59.1 (34.9)	
Duration of time since stopping PN to the time of the study, mean years (SD)	–	–	6.8 (4.3)
Weight-for-age *z* score, mean (SD)	−0.1 (0.9)	0.0 (0.9)	−0.2 (0.9)
Height-for-age *z* score, mean (SD)	−0.1 (1.1)	−0.1 (1.0)	−0.1 (1.2)
Body-mass-index *z* score, mean (SD)	−0.1 (1.0)	0.2 (0.9)	−0.3 (1.0)
Bone mineral density, total body less head *z* score as determined by DXA, mean (SD)	−0.5 (1.2)	−0.4 (1.3)	−0.6 (1.1)

Abbreviations: DXA, dual energy x-ray absorptiometry; PN, parenteral nutrition; PNDI, parenteral nutrition dependency index.

Several techniques exist to analyze body composition. Each method varies in complexity and ease of use, and each is governed by assumptions that may affect their suitability for different populations.[34] BIA is noninvasive and inexpensive and can be applied to pediatric patients at the bedside to derive clinically relevant information. However, it does assume a constant hydration state. Therefore, altered fluid status can significantly influence the accuracy of impedance measurements and special consideration must be taken.[24] In a study from Yap et al, it was found that in patients receiving intravenous fluids, BIA measurements are significantly affected by the ionic content of parenterally infused solutions for ≥2 h after infusion. These findings suggest that BIA is detecting the electroionic status of body fluids.[24] Therefore, controlling for fluid intake is of utmost importance when using this method.

The validity of BIA in patients with IF has been investigated in the past.[25] Specifically, BIA was compared to deuterium dilution in measuring total body water and lean body mass in a relatively small sample of 15 children. The results concluded that body composition measurements by BIA method were in agreement with isotope dilution, with a small mean bias of 1.7% for %FM. Although BIA was stated to be a reliable method for body composition assessment in patients with IF from that study, the limits of agreement were wide

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TABLE 2 Comparison of body composition assessment methods in children with intestinal failure (percentage, $n = 68$).

	Body composition output data mean (SD)	Mean bias % (SD) alternative method vs DXA	95% CI alternative method vs DXA	*P* value[a]
Percentage of fat mass				
DXA	25.7 (6.6)	–	–	–
BIA	24.7 (3.7)	0.69 (4.8)	(−0.5, 1.9)	0.26
Skinfold	23.2 (8.0)	2.47 (5.9)	(1.0, 4.0)	0.001
Percentage of fat-free mass				
DXA	74.3 (6.6)	–	–	–
BIA	75.3 (3.7)	−0.69 (4.8)	(−1.9, 0.5)	0.26
Skinfold	76.8 (8.0)	−2.47 (5.9)	(−4.0, −1.0)	0.001

Note: All values are mean with SD.

Abbreviations: BIA, bioelectrical impedance analysis; DXA, dual energy x-ray absorptiometry.

[a]Method compared to DXA (paired *t* test).

TABLE 3 Comparison of body composition assessment methods in children with intestinal failure (kilograms, $n = 68$).

	Body composition output data mean (SD)	Mean bias kilograms (SD) alternative method vs DXA	95% CI alternative method vs DXA	*P* value[a]
Fat mass (kg)				
DXA	8.5 (5.6)	–	–	–
BIA	8.0 (5.0)	0.14 (1.9)	(−0.4 to 0.6)	0.59
Fat-free mass (kg)				
DXA	23.8 (6.5)	–	–	–
BIA	23.2 (11.8)	−0.13 (1.9)	(−0.6 to 0.4)	0.59

Note: All values are mean with SD.

Abbreviations: BIA, bioelectrical impedance analysis; DXA, dual energy x-ray absorptiometry.

[a]Method compared to DXA (paired *t* test).

(−16.9% to 20.3%), suggesting that individual BIA results must be interpreted with caution. Importantly, fluid intake (enteral and/or parenteral) was not controlled for in that study. This would have an impact on results given that the fluid and electrolyte content directly affects the measured reactants.[24] The administration of the deuterium enterally in that study poses a challenge in the interpretation of the results given that children with IF have underlying malabsorption.

In a study of adults with short bowel syndrome, BIA estimates of FFM were examined for agreement with DXA.[23] Mean bias for FFM by the two techniques was −1.6 kg. However, the limits of agreement were wide (−10.7 to +7.4 kg). Fluid shifts and intravenous fluid administration prior to the study procedures were thought to have influenced the hydration state and hence the accuracy of BIA estimates in this study.[23]

Many BIA devices have equations developed by manufacturers for their specific instrument that are undisclosed by the company.[35] These equations are largely based on adult populations. Accuracy of the prediction equations depends on numerous assumptions, including constant tissue hydration and homogenous conduction of the electrical current through a cylindrical body. The former assumption is poorly upheld in childhood as body water compartments and the FFM hydration are undergoing rapid changes that affect fluid distribution and electrolyte concentrations between the intracellular and extracellular compartments. Many of the equations do not apply age-specific hydration factors.[36,37] A study by Fomon et al showed that the total body water decreased from 80.6% of FFM at birth to 77% at age 10 years.[31,38] In adults, the hydration of FFM is estimated to be 73%. Hydration factors should therefore be age-specific, particularly in children. When validating BIA equations, the changes in hydration ratio that occur throughout childhood must also be accounted for.[39] Fjeld et al developed predictive equations from BIA measurements specific to well and malnourished children in which they cross-validated measurement of total body water against oxygen-18 isotope dilution ($r = 0.99$).[30] These equations were used in our study under the assumption that the hydration of FFM is 77%.[31] Both of these assumptions may have contributed to the significant effect of age and sex on the difference between BIA and DXA in the current study. Nevertheless, no predictive equations specific for the IF population currently exist. Further, the mean age of the sample was around 9 years, and there were few patients who were younger than 3 years or older than 13 years, in which different hydration factors may be more appropriate. In assessing different hydration factors for FFM, different hydration factors based on age were attempted and there was no significant change in bias in the results. In addition to assumption error, body composition assessments are affected by measurement error. The effect of age may also be explained by greater movement in the younger children, which can significantly effect BIA results. Older children were able to stay still longer and understand instruction better than the younger children.

In our current study, body composition determined by skinfold measurements were not comparable to DXA. Larger mean bias and wider 95% CI was seen between the four-point skinfold technique and DXA such that the results obtained from the skinfold technique were significantly different from DXA. Based on these results, skinfolds cannot be used as an alternative to measuring body composition in patients with IF. Despite this, skinfold thickness measurements have advantage of being readily available, inexpensive, and safe.[40] The accuracy of skinfold measurements is influenced by the operator experience/skill, location of the measurement, patient cooperation, and the prediction equation used to derive percentage of FM.[40] Body composition prediction using skinfolds are useful alternatives to advanced techniques, but the results of our study suggest that four-point skinfold thickness measurements are not an alternative to DXA to obtain assessments of body composition in children with IF. The skinfold method can be an option for many clinicians to understand growth quality when other body composition

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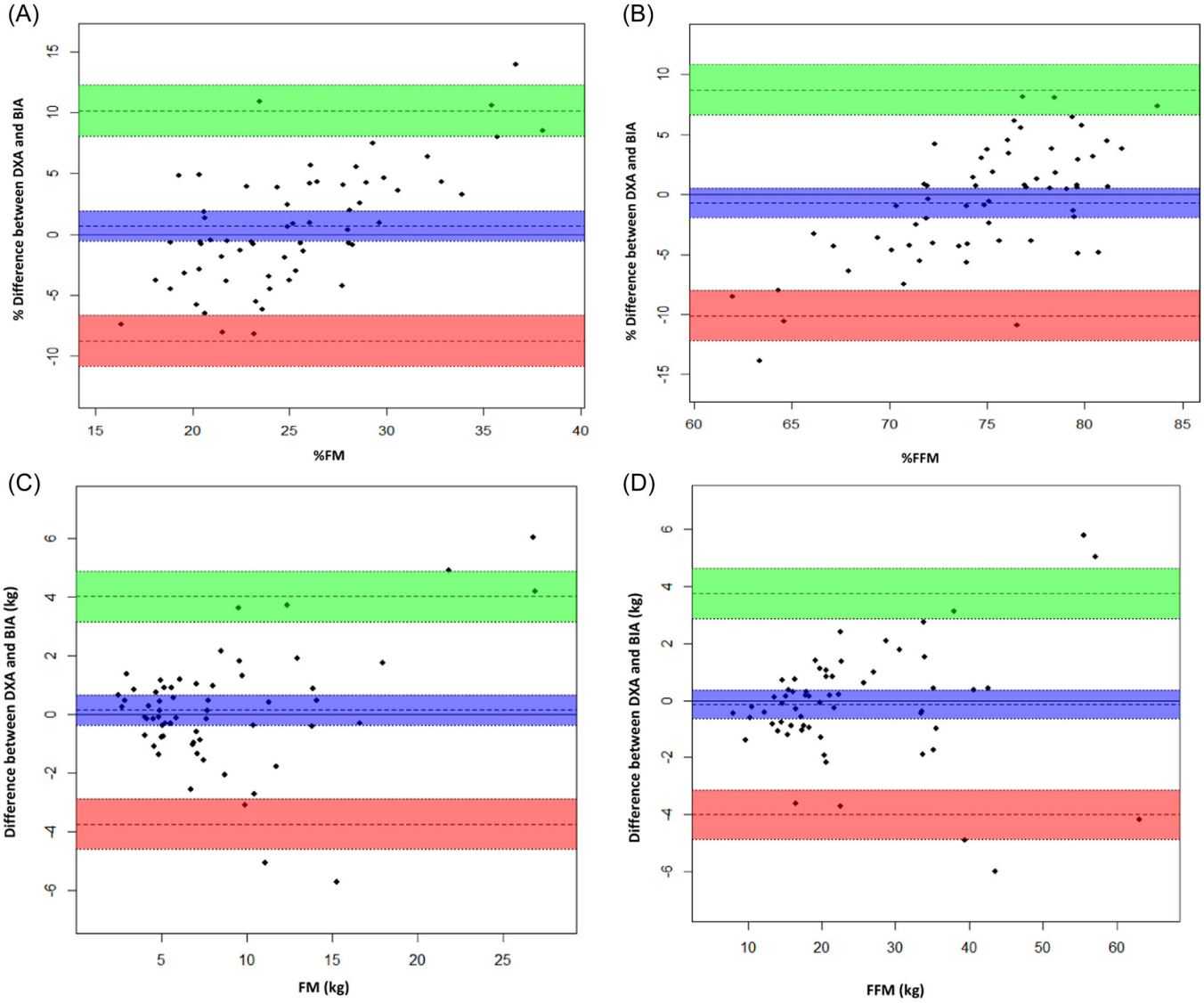

FIGURE 1 (A) Bland-Altman plot for agreement between BIA and DXA for %FM in children with intestinal failure ($n = 68$). The middle region represents the 95% CI (−0.5 to 1.9). The dotted line within the middle region (----) represents mean bias (0.69%). The upper and lower regions represent upper and lower limit of agreement between DXA and BIA. (B) Bland-Altman plot for agreement between BIA and DXA for %FFM in children with intestinal failure ($n = 68$). The middle region represents the 95% CI (−1.9% to 0.5%). The dotted line within the middle region (----) represents mean bias (−0.69%). The upper and lower regions represent upper and lower limit of agreement between DXA and BIA. (C) Bland-Altman plot for agreement between BIA and DXA for FM (kilograms) in children with intestinal failure ($n = 68$). The middle region represents the 95% CI (−0.6 to 0.4 kg). The dotted line within the middle region (----) represents mean bias (0.14 kg). The upper and lower regions represent upper and lower limit of agreement between DXA and BIA. (D) Bland-Altman plot for agreement between BIA and DXA for FFM (kilograms) in children with intestinal failure ($n = 68$). The middle region represents the 95% CI (−0.5 to 0.6 kg). The dotted line within the middle region (----) represents mean bias (−0.13 kg). The upper and lower regions represent upper and lower limit of agreement between DXA and BIA. BIA, bioelectrical impedance analysis; DXA, dual energy x-ray absorptiometry.

methods are not available. In the pediatric population, rather than using the four-point skinfold measurements, the individual values of each skinfold measurement or the mid-upper arm circumference can be converted to percentiles. These can then be compared with reference norms to help guide clinical decisions regarding nutrition support in patients with IF.[41]

Despite BIA being noninvasive, inexpensive, and comparable to DXA as per our study findings, the issue of practicality remains. There is significant preparation in regards to fasting that is required to control for the effect of fluid to complete an accurate BIA measurement. This can often be difficult to achieve, particularly in younger children, those who are PN-dependent, and hospitalized children. In younger children, they often have more frequent feeding regimens and may have a harder time refraining from food/fluid for several hours. Children who are PN-dependent need to be cycled off of their PN for a minimum of

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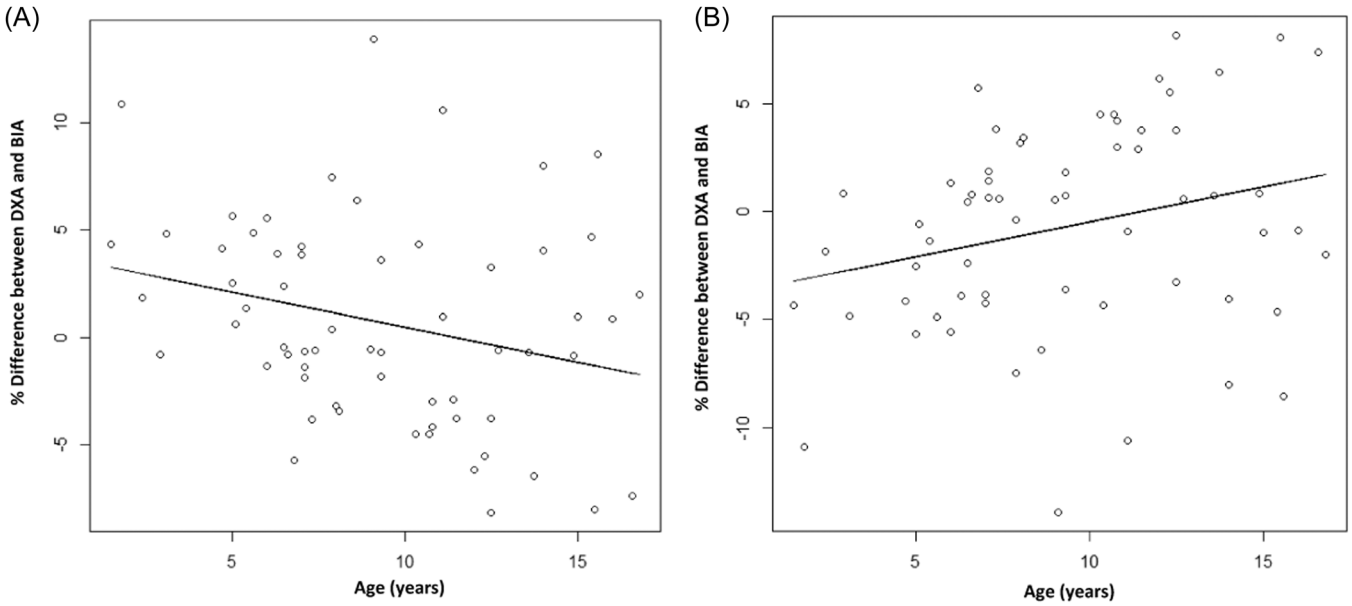

FIGURE 2 (A) Effect of age on the difference between DXA and BIA for %FM in children with intestinal failure ($n = 68$). Marginal linear mixed model revealed a significant negative association between age and the difference between DXA and BIA for %FM ($r^2 = 0.06$, $P = 0.04$). Regression equation is $y = 3.73 + -0.33$ (age). (B) Effect of age on the difference between DXA and BIA for %FFM in children with intestinal failure ($n = 68$). Marginal linear mixed model revealed a significant positive association between age and the difference between DXA and BIA for %FFM ($r^2 = 0.06$, $P = 0.04$). Regression equation is $y = -3.73 + 0.33$ (age). BIA, bioelectrical impedance analysis; DXA, dual energy x-ray absorptiometry.

4 h prior to BIA mesaurement, which may not always be feasible given their PN schedule. As for hospitalized patients, they often require continuous intravenous support, which does not allow for the fasting period. Extra planning and coordination is required when using BIA in children with IF.

Our study demonstrated good agreement between BIA and DXA without significant effect of feeding method (PN-supported vs fully enterally fed). This suggests appropriate control for fluid intake, both intravenous and enteral. The fact that age and sex demonstrated significant effects suggests that age- and sex-specific hydration of FFM variables should be used when converting total body water to %FFM. Further work is required to determine if results are consistent at the individual level and if BIA is sensitive enough to capture small changes to body composition over short time periods. The development of BIA equations specific to the IF population should also be explored.

CONCLUCISON

This is the largest study to date assessing BIA against DXA as a tool to measure body composition in pediatric IF. This work suggests BIA is an acceptable clinical tool for assessing body composition in a cohort of pediatric patients with IF. BIA is a simple, more efficient, and cost-effective method for routine assessment of body composition in clinical practice. However, special consideration is required to control for the effect of hydration and electrolytes on BIA. Therefore, measurement in individual patients who are hospitalized could pose a challenge. Future studies are needed to assess the efficacy of BIA in capturing changes in body composition within a short time frame to further support its′ routine use at the individual level in the clinical setting, particularly in patients who are followed on an outpatient basis. Clinical assessment in this area can lead to refinement and personalization of nutrition care plans and optimize timing of physiotherapy supportive care interventions. Thus, advancement in this area may have a long-lasting impact on patient quality of life and successful outcomes.

AUTHOR CONTRIBUTIONS

Dianna Yanchis and Glenda Courtney-Martin equally contributed to the conception and design of the research; Dianna Yanchis, Glenda Courtney-Martin, Stephanie So, Catherine Patterson, Christina Belza, Sylvia Wong-Sterling, Elizabeth Garofalo, Carina Silva, YaronAvitzur, Jessie Hulst, and Paul W. Wales contributed to patient recruitment and data collection; Dianna Yanchis, Glenda Courtney-Martin, Dehan Kong, and Libai Xuyx contributed to the data analysis; Dianna Yanchis, Glenda Courtney-Martin, Dehan Kong, and Libai Xuyx contributed to the interpretation of the data; and Dianna Yanchis and Glenda Courtney-Martin drafted the manuscript. All authors critically revised the manuscript, agree to be fully accountable for ensuring the integrity and accuracy of the work, and read and approved the final manuscript. Glenda Courntey-Martin is responsible for the final contents of the manuscript.

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CONFLICT OF INTEREST STATEMENT

The authors declare no conflict of interest.

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How to cite this article: Yanchis D, So S, Patterson C, et al. Assessment of body composition in pediatric intestinal failure: a comparison study. *J Parenter Enteral Nutr*. 2023;47: 920-929. doi:10.1002/jpen.2540

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